

Problems on Tuple

Each problem includes a sample input and expected output for better understanding.

SECTION A - BASIC

Core syntax, built-in operations, and straightforward practice.

1. Write a Python program to create a tuple and print its elements.

Input	(10, 20, 30, 40)
Expected Output	(10, 20, 30, 40)

2. Write a Python program to find the length of a tuple.

Input	(1, 2, 3, 4, 5)
Expected Output	Length = 5

3. Write a Python program to count the occurrences of an element in a tuple.

Input	Tuple = (1, 2, 2, 3, 2, 4) Element = 2
Expected Output	Occurrences of 2 = 3

4. Write a Python program to find the index of an element in a tuple.

Input	Tuple = (10, 20, 30, 40) Element = 30
Expected Output	Index = 2

5. Write a Python program to convert a tuple into a list and a list into a tuple.

Input	Tuple = (1, 2, 3) List = [4, 5, 6]
Expected Output	Tuple to List = [1, 2, 3] List to Tuple = (4, 5, 6)

6. Write a Python program to find the maximum and minimum values in a tuple.

Input	(8, 3, 15, 1, 9)
Expected Output	Maximum = 15, Minimum = 1

7. Write a Python program to slice a tuple and print specific elements.

Input	Tuple = (10, 20, 30, 40, 50) Slice = [1:4]
Expected Output	(20, 30, 40)

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8. Write a Python program to concatenate two tuples.

Input	Tuple 1 = (1, 2, 3) Tuple 2 = (4, 5)
Expected Output	(1, 2, 3, 4, 5)

9. Write a Python program to check whether an element exists in a tuple.

Input	Tuple = (10, 20, 30) Element = 20
Expected Output	20 exists in the tuple.

10. Write a Python program to create a tuple with repeated elements.

Input	Element = 5 Repeat = 4
Expected Output	(5, 5, 5, 5)

11. Write a Python program to unpack a tuple into separate variables.

Input	(101, "Amit", 85.5)
Expected Output	Roll = 101 Name = Amit Marks = 85.5

12. Write a Python program to sort tuple elements and display the result.

Input	(5, 2, 8, 1, 4)
Expected Output	(1, 2, 4, 5, 8)

13. Write a Python program to remove duplicate values from a tuple.

Input	(1, 2, 2, 3, 1, 4)
Expected Output	(1, 2, 3, 4)

14. Write a Python program to create a nested tuple and access its inner elements.

Input	(1, 2, (10, 20, 30))
Expected Output	Inner tuple = (10, 20, 30) Second inner element = 20

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15. Write a Python program to swap two tuples.

Input	Tuple A = (1, 2) Tuple B = (3, 4)
Expected Output	Tuple A = (3, 4) Tuple B = (1, 2)

16. Write a Python program to find the sum of numeric elements in a tuple.

Input	(10, 20, 30, 40)
Expected Output	Sum = 100

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SECTION B - INTERMEDIATE

Multi-step logic and practical manipulation of the topic.

1. Write a Python program to reverse the elements of a tuple using slicing.

Input	(1, 2, 3, 4)
Expected Output	(4, 3, 2, 1)

2. Write a Python program to calculate the element-wise sum of two tuples.

Input	Tuple 1 = (1, 2, 3) Tuple 2 = (4, 5, 6)
Expected Output	(5, 7, 9)

3. Write a Python program to find all values that occur more than once in a tuple.

Input	(1, 2, 2, 3, 4, 4, 4)
Expected Output	Repeated values = (2, 4)

4. Write a Python program to sort a list of tuples by the second element of each tuple.

Input	[('A', 75), ('B', 92), ('C', 80)]
Expected Output	[('A', 75), ('C', 80), ('B', 92)]

5. Write a Python program to unzip a tuple of pairs into two separate tuples.

Input	((1, 'A'), (2, 'B'), (3, 'C'))
Expected Output	Numbers = (1, 2, 3) Letters = ('A', 'B', 'C')

6. Write a Python program to unpack a tuple using a starred variable for the middle elements.

Input	(10, 20, 30, 40, 50)
Expected Output	First = 10 Middle = (20, 30, 40) Last = 50

7. Write a Python program to calculate column-wise sums of a tuple-based matrix.

Input	((1, 2, 3), (4, 5, 6), (7, 8, 9))
Expected Output	Column sums = (12, 15, 18)

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8. Write a Python program to create a new tuple containing only values greater than a given threshold.

Input	Tuple = (5, 12, 3, 18, 7) Threshold = 10
Expected Output	(12, 18)

9. Write a Python program to interleave two tuples of equal length.

Input	Tuple 1 = (1, 3, 5) Tuple 2 = (2, 4, 6)
Expected Output	(1, 2, 3, 4, 5, 6)

10. Write a Python program to create a tuple of (value, frequency) pairs for all distinct values.

Input	(1, 1, 2, 3, 3, 3)
Expected Output	((1, 2), (2, 1), (3, 3))

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SECTION C - ADVANCED

Algorithmic problem solving and deeper application of the topic.

1. Write a Python program to transpose a matrix represented as a tuple of tuples.

Input	((1, 2, 3), (4, 5, 6))
Expected Output	((1, 4), (2, 5), (3, 6))

2. Write a Python program to rotate a tuple-based matrix by 90 degrees clockwise.

Input	((1, 2, 3), (4, 5, 6))
Expected Output	((4, 1), (5, 2), (6, 3))

3. Write a Python program to generate the Cartesian product of two tuples.

Input	Tuple 1 = (1, 2) Tuple 2 = ('A', 'B')
Expected Output	((1, 'A'), (1, 'B'), (2, 'A'), (2, 'B'))

4. Write a Python program to generate all 2-element combinations from a tuple.

Input	(1, 2, 3, 4)
Expected Output	((1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4))

5. Write a Python program to recursively flatten a nested tuple into a single tuple.

Input	(1, (2, (3, 4)), 5)
Expected Output	(1, 2, 3, 4, 5)

6. Write a Python program to sort student record tuples first by marks and then by name.

Input	((101, 'Amit', 85), (102, 'Neha', 92), (103, 'Ravi', 85))
Expected Output	((101, 'Amit', 85), (103, 'Ravi', 85), (102, 'Neha', 92))

7. Write a Python program to group tuple records by a selected field.

Input	((('Amit', 'DS'), ('Neha', 'AI'), ('Ravi', 'DS')))
Expected Output	DS = ('Amit', 'Ravi') AI = ('Neha',)

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8. Write a Python program to perform a circular left rotation of a tuple by n positions.

Input	Tuple = (1, 2, 3, 4, 5) n = 2
Expected Output	(3, 4, 5, 1, 2)

9. Write a Python program to find the longest increasing contiguous subsequence in a tuple.

Input	(1, 2, 5, 3, 4, 6, 7, 2)
Expected Output	Subsequence = (3, 4, 6, 7) Length = 4

10. Write a Python program to merge two sorted tuples into one sorted tuple without using sort().

Input	Tuple 1 = (1, 4, 7) Tuple 2 = (2, 3, 8, 9)
Expected Output	(1, 2, 3, 4, 7, 8, 9)